

FROM 0% TO 90%

Systematic RAG Retrieval Optimization

A case study in data-driven engineering: How 8 focused sessions and 7.5 hours of systematic work transformed a broken search system into elite-tier accuracy.

90%

ACCURACY

8

SESSIONS

100%

ARCHITECTURAL

100%

COMPARATIVE

THE SETUP: PHASE 7 PRODUCTION RAG

WHAT WE BUILT

Production RAG system for homelab documentation:
Streamlit chat UI, 611 embedded chunks across 85 documents, semantic caching with 200× speedup, SQLite conversation persistence.

✓ WHAT WORKED

- Fast queries (2.5s avg)
- 200× cache speedup
- Full observability

⚠ WHAT DIDN'T

- Similarity plateau: 0.60
- Fallback rate: 70%
- Wrong docs returned



The Brutal Truth: Created 20-query test suite → Discovered 0% factual accuracy Query: "What is Nomad?" → Returned session notes instead of ADR-001 definition

The system worked... but gave wrong answers 100% of the time for factual queries

ROOT CAUSE ANALYSIS

THE FUNDAMENTAL DIAGNOSIS

Analysis identified three distinct technical failures preventing accurate retrieval. Vector search alone finds "related" noise instead of "authoritative" signals.



KEYWORDS

Vector-only search misses exact term matches. Query: "What is Nomad?" result in semantic drift. Semantic $\neq$ Exact.



AUTHORITY

Session notes rank equally with authoritative ADRs. Frequency of mentions outranked structural definitions.



TOPIC MATCH

Docs that mention a topic outrank docs primary ABOUT a topic. The "About" signal was buried in chunk frequency.

THE THREE PILLARS STRATEGY

Systematic optimization through three complementary techniques, validated with a 20-query test suite. Every pillar addresses a root cause.



PILLAR 1: HYBRID

BM25 (Keywords) + Vector (Semantics). Fixed the exact-match failure.

Result: 0% → 20%



PILLAR 2: AUTHORITY

ADR boosting (1.5×) and noise de-boosting (0.8×). Rank definitions first.

Result: 20% → 75%



PILLAR 3: TOPIC

Title and filename matching boost (1.5×). Ensure "About X" outranks "Mentions X".

Result: 75% → 90%

BREAKTHROUGH #1: HYBRID SEARCH

THE TECHNICAL FORMULA

$$\text{Score} = 0.3 \times \text{BM25} + 0.7 \times \text{Vector}$$

Custom tokenizer preserves technical identifiers like llama3.1:8b and ADR-001 as single atoms rather than split fragments.

THE OUTCOME

- Vector Only: "What is Nomad?" finds casual session mentions of the cluster.
- Hybrid: Exact "Nomad" keyword forces the definition ADR to the top rank.

Foundational Validation Complete.

BREAKTHROUGH #2: DOCUMENT AUTHORITY

Insight: Not all documents are equal. ADRs are authoritative definitions; Session notes are casual noise. We measured the gap and implemented data-driven boosting.

BOOST MULTIPLIERS

ADR: 1.5× (Definition)

Reports: 1.3× (Formal)

Sessions: 0.8× (De-boost)

THE 55% JUMP

Gap analysis showed ADR-001 needed exactly 1.5× to clear the noise. Calculated, not guessed.

Result: 20% → 75%

BREAKTHROUGH #3: TOPIC AUTHORITY

SOLUTION: TITLE MATCHING

Resolved "Semantic Competition" where multiple docs mention a query term. Boost documents where terms appear in the TITLE or FILENAME.

BEFORE (MENTION > SUBJECT)

Query: "What is Traefik?"

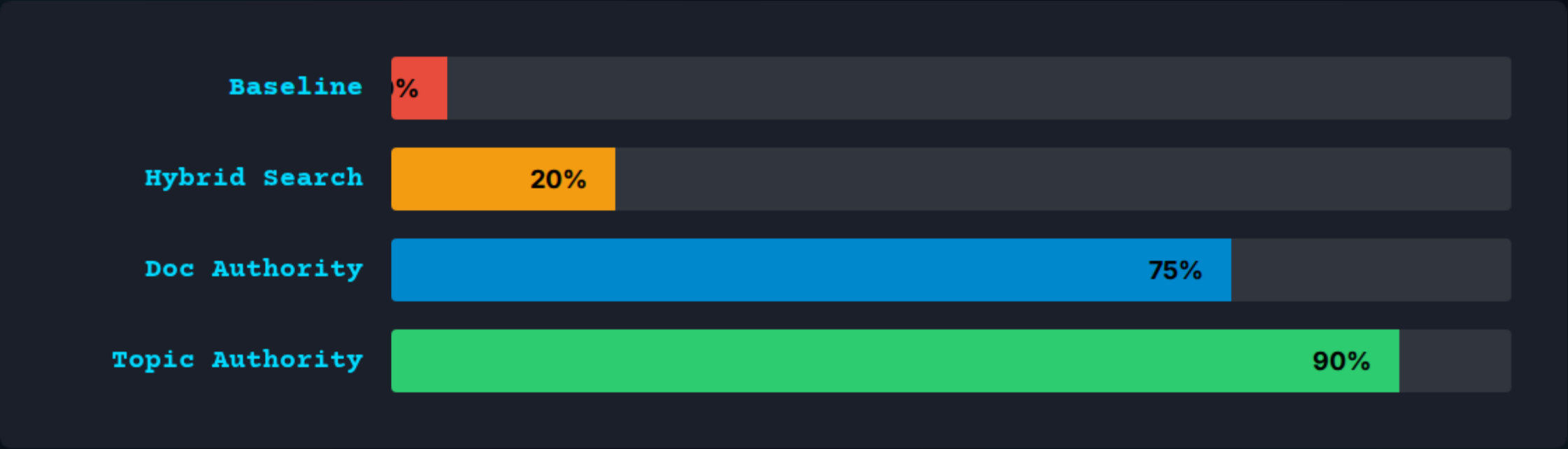
- ✗ ADR-002 mentions it 50 times (Rank 1)
- ✓ ADR-003 is ABOUT it (Rank 4)

AFTER (SUBJECT > MENTION)

- ADR-003 contains "Traefik" in title.
 - Receives 1.5× Topic Boost.
 - Rises to Rank #1.

RESULT: 75% → 90% accuracy across all categories.

THE TRANSFORMATION



FACTUAL 80% Correct	HOW-TO 80% Correct	ARCHITECTURAL 100% Perfect	COMPARATIVE 100% Perfect
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WORKFLOW DISCOVERY

INSIGHT: TOOL SELECTION MULTIPLIER

Session 6 revealed that matching the AI interface to the task tripled productivity. Different tools for different development stages.

CLAUDE CHAT

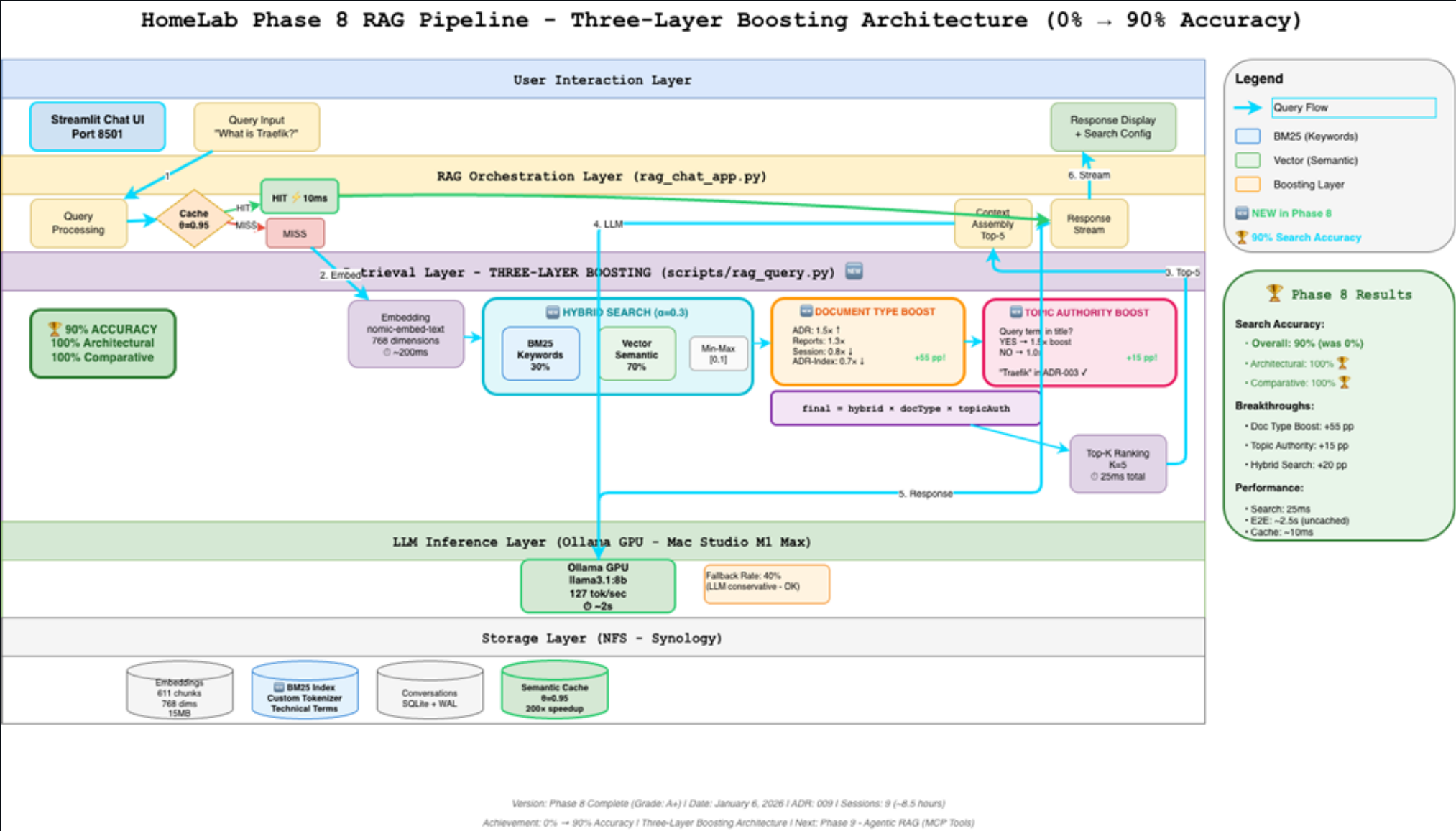
Best for planning, strategy, and context preservation.
Core for architectural decision-making and session logging.

CLAUDE CODE

Best for implementation, test validation, and git operations. Used for rapid re-ranking logic iteration.

Result: Topic Authority implemented in 20 minutes with zero regressions and perfect git history.

TECHNICAL ARCHITECTURE (1/2)



TECHNICAL ARCHITECTURE (2/2)

$$\text{Final Score} = (\text{Hybrid_Rank}) \times (\text{Doc_Type_Boost}) \times (\text{Topic_Match_Boost})$$



RETRIEVAL

Hybrid BM25 + Vector Search ($\alpha=0.3$). Min-Max Normalization ensures scores are mathematically comparable.



HEURISTIC LAYER

Multiplicative boosting for ADR types and Title matching. Heuristics solve what semantic math cannot reach.



LATENCY

Search Latency: 25ms average.
ADR Boost: 1.5× | Notes De-boost: 0.8× | Topic Match: 1.5×.

KEY ENGINEERING PRINCIPLES

1. MEASURE FIRST

Baseline: 0%. Diagnostic scripts revealed ADRs buried at rank #10. Every incremental change was validated against the test suite.

2. VALIDATE ASSUMPTIONS

Discovered chunk size was 10× larger than assumed (1847 chars). This discovery saved 4 hours of potential rework.

3. SIMPLE SOLUTIONS WIN

Topic Authority: 10 lines of code gave +15 percentage points. Elegance in logic beats complexity in math.

4. METHODOLOGY > MAGIC

Three pillars strategy. Each pillar solved one specific technical failure through rigour rather than luck.

90%

PHASE 8 READY

THE AGENTIC SHIFT

VISION FOR PHASE 9

From a static Knowledge Base to an Intelligent Operations Assistant.

- Live System Access: APIs for Nomad, Consul, and Prometheus.
- Intelligent Diagnosis: Knowledge Base vs. Live Cluster
- State Cross-Reference: Automate documented threshold compliance checking.

"The breakthrough wasn't better search. It was better ranking. Document authority matters more than pure similarity."

— Core Lesson from Phase 8

COMPOUND GAINS

Hybrid (+20), Doc Auth (+55), Topic Auth (+15). A multiplicative engineering journey.

SIMPLE IMPACT

Total boosting logic: <60 lines of code. Total Accuracy gain: 90 percentage points.

THE IMPACT

90%

ACCURAC
Y

100%

ARCH /
COMP

7.5

HOURS
INVESTED

"Engineering methodology transformed a broken search system
into elite-tier performance."

PHASE 8 COMPLETE | JANUARY 2026

APPENDIX

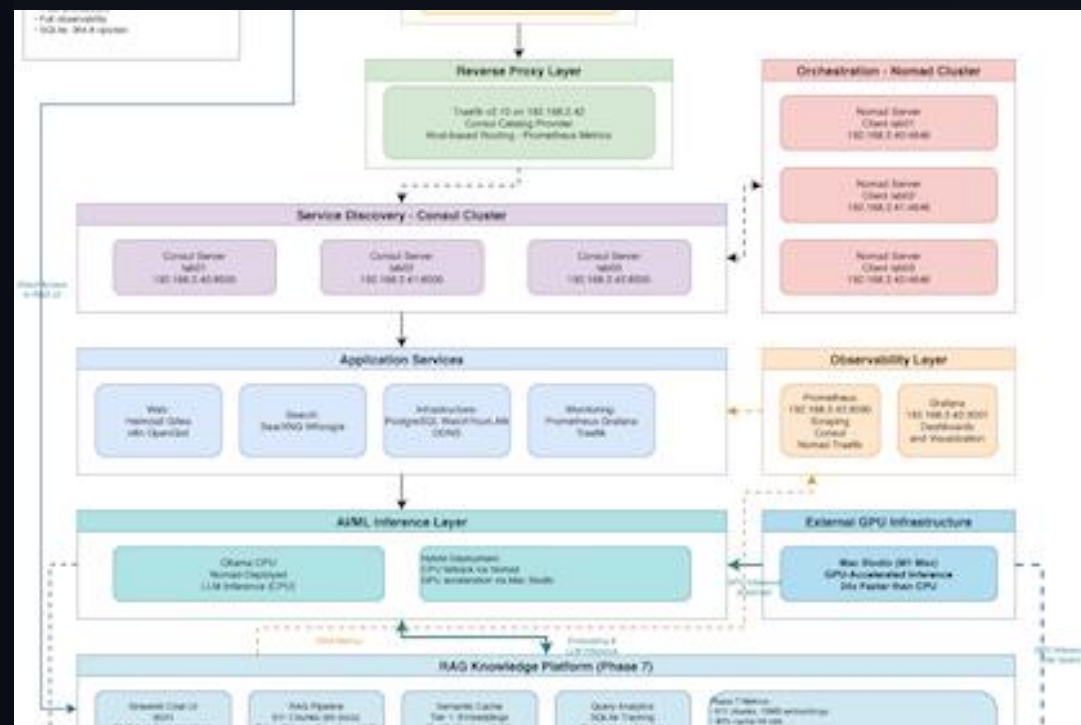
Detailed Architectural Diagrams

NOMAD CLUSTER ARCHITECTURE

SYSTEM TOPOLOGY

The Homelab environment operates on a multi-node cluster managed by HashiCorp Nomad. It ensures high availability for our search engine, storage backends, and UI components.

- Server Nodes: Governance & Scheduling
- Client Nodes: Workload Execution
- Consul Integration: Service Discovery



RAG PIPELINE ARCHITECTURE

THE RETRIEVAL FLOW

Our optimized pipeline integrates three distinct signal processing steps. It transforms raw vector similarity into authoritative context for the LLM generation step.

- Ingestion: 1847 char chunks
- Retrieval: Hybrid Alpha 0.3
- Re-Ranking: Authority + Topic Boost

